



BAULKHAM HILLS HIGH SCHOOL

2012
YEAR 11 YEARLY

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 1 hour and 30 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in every question
- Marks may be deducted for careless or badly arranged work
- Attempt all questions

Total marks – 60

This paper consists of TWO sections.

Section 1 – Multiple Choice 6 marks

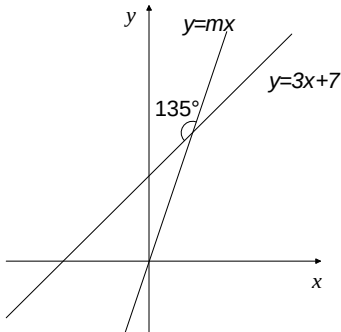
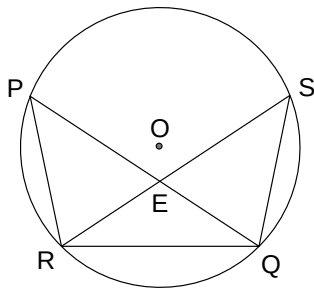
Section 2 – Extended Response 54 marks

Attempt all questions

Start a new page for each question

Section 1 –Multiple Choice (6 marks)

Attempt all questions.

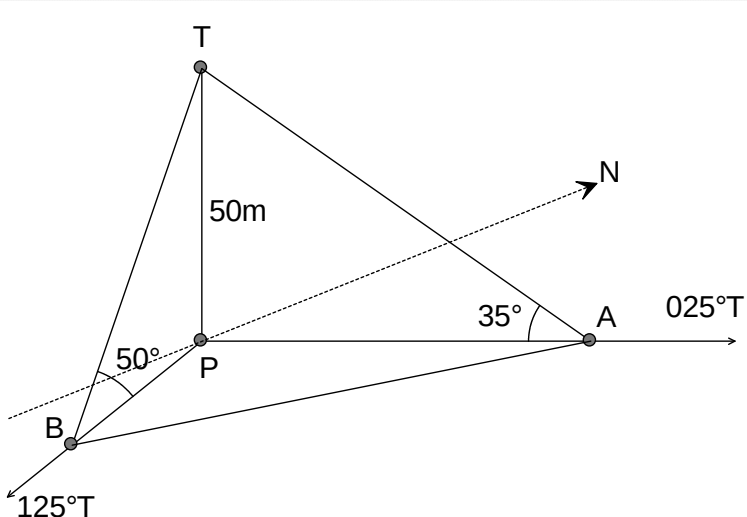
Answer the following on the answer sheet provided.		Marks
1	Let A be the point $(-2,3)$ and B be the point $(3,-4)$. Find the coordinates of the point which divides AB externally in the ratio 3: 2 (A) $(0, \frac{1}{5})$ (B) $(1, -\frac{6}{5})$ (C) $(-12,17)$ (D) $(13, -18)$	1
2	The line $y = 3x + 7$ and $y = mx$ intersect at an angle of 135°, as shown in the diagram.  A possible value for m is (A) $\frac{1}{3}$ (B) -2 (C) 3 (D) $-\frac{1}{2}$	1
3	From six girls and four boys, a committee of 3 girls and 2 boys is to be chosen. How many different committees can be formed? (A) 26 (B) 120 (C) 252 (D) 1440	1
4	Let $t = \tan \frac{\theta}{2}$ where $0 < \theta < 180^\circ$. Which of the following gives the correct expression for $\sec \theta + \tan \theta$? (A) $\frac{1-t}{1+t}$ (B) $\frac{1+2t+t^2-t^3}{1-t}$ (C) $\frac{1+t}{1-t}$ (D) $\frac{t^2-2t-1}{1+t^2}$	1
5	 O is the centre of the circle. PQ and RS are chords which intersect at E. $PR \neq SQ$. Consider the following I <i>Triangles PER and SEQ are similar</i> II <i>Triangles PQR and ERQ are similar</i> Which is true? (A) I only (B) II only (C) Both I and II (D) Neither I nor II	1
6	The equation of the normal to the parabola $x^2 = 4ay$ at the variable point $P(2ap, ap^2)$ is given by $x + py = 2ap + ap^3$ How many different values of p are there such that the normal passes through the focus of the parabola? (A) 0 (B) 1 (C) 2 (D) 3	1
End of Section I		

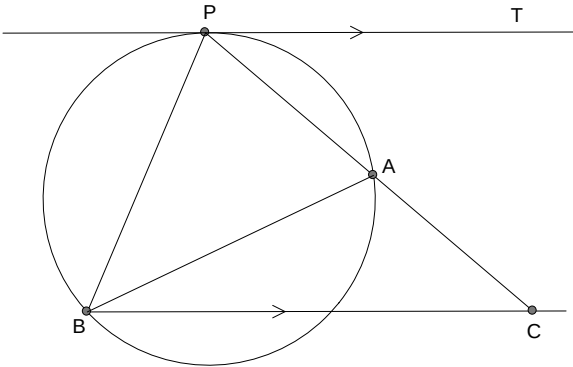
Section II – Extended Response

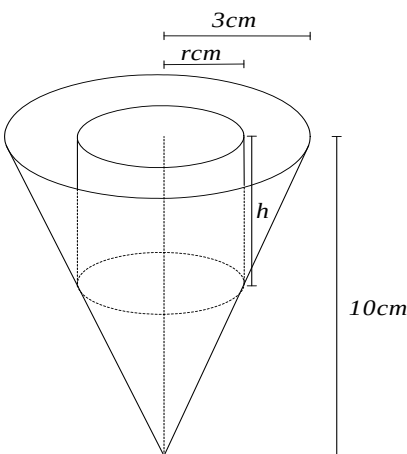
Attempt all questions. Show all necessary working.

Start each question on the appropriate page. Clearly indicate the question number.

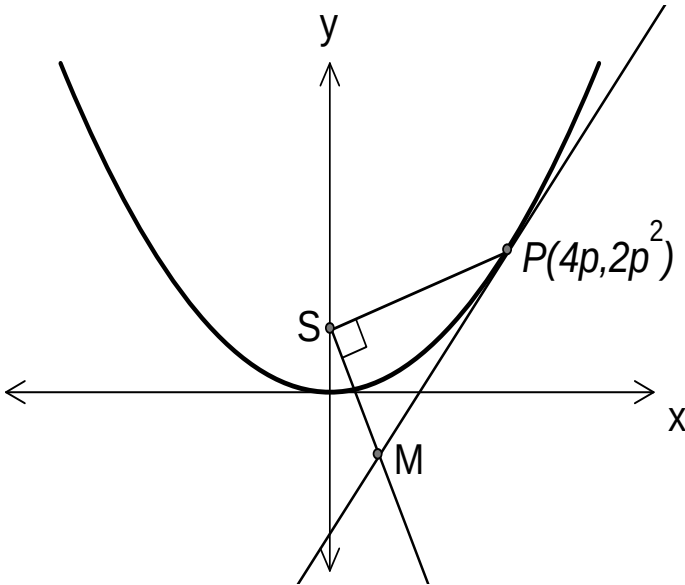
Question 7 (9 marks) - Start a new page		Marks
a)	Solve $\tan 2x = 3 \tan x$ for $0^\circ \leq x \leq 360^\circ$	3
b)	i) How many seven digit numbers can be formed using the digits 3, 3, 3, 3, 5, 5, 5 ?	1
	ii) How many five digit numbers can be formed using the digits 3, 3, 3, 3, 5, 5, 5 ?	2
c)	Solve $\frac{2x+1}{3x-2} \leq 2$	3

Question 8 (9 marks) - Start a new page		
a)	A committee of four is to be formed from 5 men and 4 women. In how many ways can the committee be chosen if :-	
	(i) there are no restrictions?	1
	(ii) it is formed with 2 men and 2 women?	1
	(iii) it contains at least 2 women?	2
b)	 PT is an observation tower $50m$ high. The bearing of two points A and B from P are $025^\circ T$ and $125^\circ T$ respectively. The angles of elevation from the points A and B to the top of the tower are 35° and 50° respectively. (i) Find $\angle APB$ (ii) Show that $AB^2 = 50^2(\cot^2 35^\circ + \cot^2 50^\circ - 2 \cot 35^\circ \cot 50^\circ \cos 100^\circ)$ (iii) Hence find AB to the nearest metre.	1 3 1

Question 9 (9 marks) - Start a new page		
a)	(i) Prove that $\sin(A + B) - \sin(A - B) = 2 \cos A \sin B$	1
	(ii) Hence or otherwise show that $2 \sin x (\cos 3x + \cos 5x + \cos 7x) = \sin 8x - \sin 2x$	2
b)	 In the diagram A, P and B are points on the circle. The line PT is tangent to the circle at P, and PA is produced to C so that BC is parallel to PT. (i) Show that $\angle PBA = \angle PCB$ (ii) Prove that triangles ABP and BCP are similar (iii) Hence deduce that $PB^2 = PA \times PC$	2 2 2

Question 10 (9 marks) - Start a new page		Marks
a)	Solve $12 \cos \theta - 5 \sin \theta + 6 = 0$, for $0^\circ \leq \theta \leq 360^\circ$, correct to the nearest degree.	3
b)	 A cylinder of radius r cm and height h cm is inscribed in a cone with base radius 3 cm and height 10 cm as in the diagram. (i) Show that the volume V of the cylinder is given by $V = \frac{10\pi r^2(3-r)}{3}$ (ii) Hence find the values of r for the cylinder which has the maximum volume. (iii) What is the maximum volume of the cylinder?	2 2 2

Question 11 (9 marks) - Start a new page		
a)	(i) By considering the difference of two cubes, show that $(2+x)^{\frac{1}{3}} - 2^{\frac{1}{3}} = \frac{x}{(2+x)^{\frac{2}{3}} + 2^{\frac{1}{3}}(2+x)^{\frac{1}{3}} + 2^{\frac{2}{3}}}$ (ii) Hence evaluate $\lim_{x \rightarrow 0} \frac{(2+x)^{\frac{1}{3}} - 2^{\frac{1}{3}}}{x}$	2
b)	(i) For the curve $y = \frac{4}{\sqrt{16+x^2}}$ find any stationary points and determine their nature. (ii) Find any points of inflexion and hence sketch the curve showing any significant features.	2 3

Question 12 (9 marks) - Start a new page		Marks
$P(4p, 2p^2)$ is a point on the parabola $x^2 = 8y$, with focus S, as in the diagram. 		
(i)	Show that the equation of the tangent at P is given by $y = px - 2p^2$	2
(ii)	Show that the equation of the line through S and perpendicular to SP is $2px + (p^2 - 1)y = 2(p^2 - 1)$	2
(iii)	The tangent and this line meet at M . Prove that the co-ordinates of M are $\left(\frac{2(p^2-1)}{p}, -2\right)$.	3
(iv)	Show that the area of $\triangle PSM = \frac{2(p^2-1)^2}{ p }$	2

End of Examination

YEAR 4 EXT 1 YEARLY MC
2012

1-D-4-C
2-B 5-A
3-B 6-B

7a) $2x = 3kx$
 $2kx = 8kx$
 $1 - kx$
 $2kx = 3kx - 3k^3x$

$kx - 3k^3x = 0$
 $kx(1 - 3k^2) = 0$
 $kx = 0$ or $kx = \frac{1}{3}$
 $x = 0, 20, 150, 150, 210, 330, 360$ ✓ all reals.

b) i) $\frac{71}{4.31} = 3.5$

ii) $33335 = x$
 $33255 = 10$
 $33555 = 10$ TOTAL = 25

c) $\frac{2x+1}{3x-2} \leq 2$ $x = \frac{2}{3}$
 $(2x+1)(3x-2) \leq 2(3x-2)^2$

$2(3x-2)^2 - (3x-2)(2x+1) \geq 0$
 $(3x-2)(6x-4-2x-1) \geq 0$
 $(3x-2)(4x-5) \geq 0$
 $2 < \frac{2}{3}$ or $x > \frac{5}{4}$

8)

a) i) $9L_4 = 126$

ii) $5L_2 \cdot 4L_2 = 60$

iii) $2W \Rightarrow 60$

$3W \rightarrow 5L_1, 4L_2 = 5 \cdot 4 = 20$

$4W \rightarrow 1$

TOTAL = $60 + 20 + 1 = 81$

b) i) $\angle APB = 125 - 25 = 100^\circ$

ii) $\angle 50^\circ = \frac{50}{BP}$

$BP = 50 \cot 50^\circ$ Similarly $AP = 50 \cot 35^\circ$

in $\triangle APB$ $AB^2 = BP^2 + AP^2 - 2BP \cdot AP \cdot \cos \angle APB$
 $= 50^2 \cot^2 50^\circ + 50^2 \cot^2 35^\circ - 2 \cdot 50^2 \cot 50^\circ \cot 35^\circ \cos 100^\circ$
 $= 50^2 (\cot^2 50^\circ + \cot^2 35^\circ - 2 \cot 50^\circ \cot 35^\circ \cos 100^\circ)$
 $= 89 \text{ m}$

9 a) i) $\ln(A+B) - \ln(A-B)$
 $= \ln A \ln B + \ln B \ln A - \ln A \ln B + \ln B \ln A$
 $= 2 \ln A \ln B = R.H.S$

ii) $2\ln 43x + 2\ln 45x + 2\ln 47x$
 $\ln(43x) = \ln(2x) + \ln(6x) = \ln(2) + \ln(6) + \ln(x)$
 $= \ln 2 + \ln 3 + \ln 2 + \ln x = 2\ln 2 + \ln 3 + \ln x$

b) i) $\angle TPA = \angle PBA$ (ANGLE FORMED BY TANGENT AND RADIUS = ANGLE IN ARC)
 $\angle TPA = \angle PCB$ (ALTERNATE $\angle$ s ON $\parallel$ LINES PT, BC)
 $\therefore \angle PBA = \angle PCB$

ii) IN $\triangle APD$ AND BCP
 $\angle PBA = \angle PCB$ (PROVED IN i)
 $\angle BPC$ IS COMMON TO BOTH TRIANGLES
 $\therefore \triangle ABP \parallel \triangle OBP$ (AA, AAA, equiangular all matching $\angle$ s =)

iii) $\frac{PB}{AP} = \frac{PC}{PB}$ (ratio of sides $\triangle P$)
 $\therefore PB^2 = PA \cdot PC$

16 a) $12\cos\theta - 5\sin\theta + 6 = 0$
 $5\sin\theta - 12\cos\theta = 6$ ($\sqrt{5^2 + 12^2} = 13$)

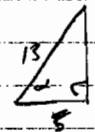
$\frac{5\sin\theta}{13} - \frac{12\cos\theta}{13} = \frac{6}{13}$

$\sin\theta \cos 2 - \cos\theta \sin 2 = \frac{6}{13}$

$\sin(\theta - 67.38) = \frac{6}{13}$

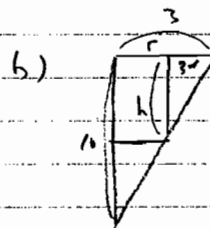
$\theta - 67.38 = 27.49$ or 152.51

$\therefore \theta = 95^\circ$ or 220°



$\theta = 67.38^\circ$

$\angle = 67.38^\circ$



$V = \pi r^2 h$

$\frac{h}{3-r} = \frac{10}{3}$

$\therefore h = \frac{10(3-r)}{3}$

Sub $\rightarrow V \therefore V = \pi r^2 \cdot \frac{10(3-r)}{3}$

$V = \frac{10\pi r^2 (3-r)}{3}$

$= \frac{10\pi}{3} (3r^2 - r^3)$

$V' = 10\pi (6r - 3r^2)$

$= 10\pi 3r(2-r)$

$= 0$ with $r = 0$ or 2

$r = 0 \Rightarrow V = 0$

$r = 2 \Rightarrow$

r	1.9	2	2.1
V'	pm	0	-m

$\therefore r = 2$

(11) a) $(A^3 - B^3) = (A - B)(A^2 + AB + B^2)$

$\therefore A - B = \frac{A^3 - B^3}{A^2 + AB + B^2}$

at $A = (2+x)^{\frac{1}{2}}$ $B = 2^{\frac{1}{2}}$

$\therefore (2+x)^{\frac{1}{2}} - 2^{\frac{1}{2}} = \frac{2+x - 2}{(2+x)^{\frac{1}{2}} + 2^{\frac{1}{2}}(2+x)^{\frac{1}{2}} + 2^{\frac{1}{2}}}$ $\xrightarrow{2=x}$ $= \frac{x}{(2+x)^{\frac{1}{2}} + 2^{\frac{1}{2}}(2+x)^{\frac{1}{2}} + 2^{\frac{1}{2}}}$

ii) $\lim_{x \rightarrow 0} \frac{(2+x)^{\frac{1}{2}} - 2^{\frac{1}{2}}}{x}$

$= \lim_{x \rightarrow 0} \frac{1}{(2+x)^{\frac{1}{2}} + 2^{\frac{1}{2}}(2+x)^{\frac{1}{2}} + 2^{\frac{1}{2}}}$

$= \frac{1}{2^{\frac{1}{2}} + 2^{\frac{1}{2}} + 2^{\frac{1}{2}}}$

$= \frac{1}{3 \cdot 2^{\frac{1}{2}}}$

11 b) i)

$y = \frac{4}{\sqrt{16+x^2}}$

$y = 4(16+x^2)^{-\frac{1}{2}}$

$y' = -2(16+x^2)^{-\frac{3}{2}} \cdot 2x$

$y' = \frac{-4x}{(16+x^2)^{\frac{3}{2}}}$

Stationary points occur when $y' = 0$

$0 = \frac{-4x}{(16+x^2)^{\frac{3}{2}}}$

$x = 0$

$\therefore$ stat. pt at $(0, 1)$

Testing $x = 0$

$x \quad -1 \quad 0 \quad 1$
 $y' \quad \frac{4}{17^{\frac{3}{2}}} \quad 0 \quad -\frac{4}{17^{\frac{3}{2}}}$

$\therefore$ max turning point at $(0, 1)$

ii) $y'' = -4(16+x^2)^{-\frac{3}{2}} - 4x \cdot -\frac{3}{2} \cdot 2x(16+x^2)^{-\frac{5}{2}}$

$= \frac{12x^2}{(16+x^2)^{\frac{5}{2}}} - \frac{4(16+x^2)^{\frac{1}{2}}}{(16+x^2)^{\frac{5}{2}}}$

$y'' = \frac{8x^2 - 64}{(16+x^2)^{\frac{5}{2}}} = \frac{8(x^2 - 8)}{(16+x^2)^{\frac{5}{2}}}$

Points of inflexion may occur when $y'' = 0$

ie $8x^2 - 64 = 0$

$x^2 = 8$

$x = \pm 2\sqrt{2}$

Testing

x	-3	$-2\sqrt{2}$	0	$2\sqrt{2}$	3
y''	$\frac{8}{125}$	0	-1	0	$\frac{+8}{125}$
	U		n		U

$\therefore$ concavity changes at $x = \pm 2\sqrt{2}$, $y = \frac{4}{\sqrt{24}}$ or $\frac{\sqrt{6}}{3}$

$\therefore$ Points of infle at $(\pm 2\sqrt{2}, \frac{\sqrt{6}}{3})$

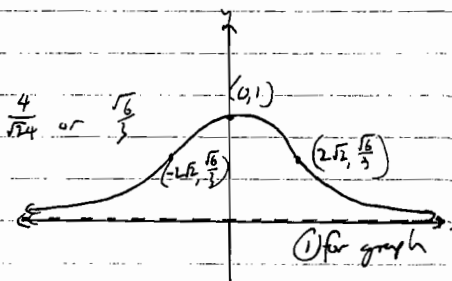
as $x \rightarrow \infty$, $y \rightarrow 0^+$, as $x \rightarrow -\infty$, $y \rightarrow 0^+$

1 for $\frac{dy}{dx}$

① for classifying stat pt.

① for y''

① for P.O.I.



12

(42)

$$z^2 = 84$$

$$y = \frac{z^2}{p}$$

$$y' = \frac{2z}{p} = \frac{z}{p}$$

$$= p \text{ at } z = 4p$$

$$\text{or } \frac{dy}{dx} = \frac{dz}{dt} \cdot \frac{dt}{dx}$$

$$= y' \cdot \frac{dt}{dx}$$

$$= t$$

$$= p \text{ at } x = 4p$$

*

$$\text{equation of tangent is } y - zp^2 = p(z - 4p)$$

$$y = px - 4p^2 + zp^2$$

$$y = px - 2p^2$$

$$-x \quad \perp$$

*

$$\text{gradient of } SP = \frac{1}{p}$$

$$\therefore \text{line is } y = \frac{1}{p}x - \frac{1}{p}$$

$$\text{gradient of } SP = -\left(\frac{4p}{2p^2 - 2}\right)$$

$$= \frac{-2p}{p^2 - 1}$$

$$-1$$

*

$$\therefore \text{equation of } SP \text{ is } y - 2 = \frac{-2p}{p^2 - 1}(x - 0)$$

$$(p^2 - 1)y - 2(p^2 - 1) = -2px + 4p$$

$$-1$$

*

$$2px + (p^2 - 1)y = 2p^2 + 2(p^2 - 1)$$

$$-x$$

$$y = px - 2p^2 \quad \text{--- (1)}$$

$$2px + (p^2 - 1)y = 2(p^2 - 1) \quad \text{--- (2)}$$

$$\text{from (1) substit } px = 8 + 2p^2 \Rightarrow (2)$$

$$\perp$$

*

$$y + 4p^2 + p^2y - y = 2p^2 - 2$$

$$2y + 4p^2 + p^2y - y = 2p^2 - 2$$

$$(p^2 + 1)y = -2 - 2p^2$$

$$y = \frac{-2(1 + p^2)}{p^2 + 1} = -2$$

$$\perp$$

*

$$\text{subst } y = -2$$

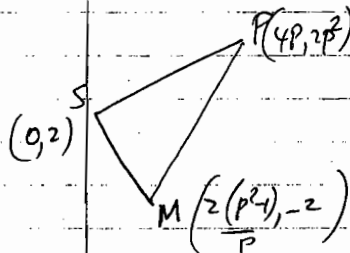
$$\therefore px = -2 + 2p^2$$

$$x = \frac{2(p^2 - 1)}{p}$$

$$\therefore M \text{ has co-ord } \left(\frac{2(p^2 - 1)}{p}, -2 \right)$$

$$-1$$

*



$$\text{Area of } \Delta = \frac{1}{2} SP \cdot SM$$

$$= \frac{1}{2} \sqrt{SP^2} \sqrt{SM^2}$$

$$= \frac{1}{2} \sqrt{(4p)^2 + (2p^2 - 2)^2} \sqrt{\left(\frac{2(p^2 - 1)}{p}\right)^2 + 16}$$

$$= \frac{1}{2} \sqrt{16p^2 + 4p^4 - 8p^2 + 4} \sqrt{\frac{4p^4 - 8p^2 + 4 + 16p^2}{p^2}}$$

$$= \frac{1}{2} \sqrt{4p^4 + 8p^2 + 4} \sqrt{\frac{4p^4 - 8p^2 + 4 + 16p^2}{p^2}}$$

$$= \frac{1}{2} \sqrt{4(p^2 + 1)^2} \sqrt{\frac{4(p^2 + 1)^2}{p^2}}$$

$$= \frac{1}{2} \cdot \frac{4(p^2 + 1)^2}{|p|}$$

$$= \frac{2(p^2 + 1)^2}{|p|}$$

$$\perp$$